STUDENT (ID, Name, LastName, Email, City, ...)



1.7K blocks, 150K tuples

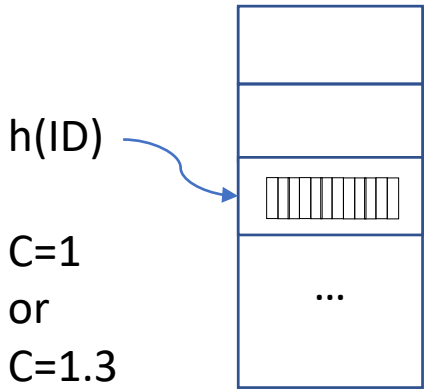
```
SELECT *
  FROM STUDENT
  WHERE City="Milan
```

Cost =

```
SELECT *
  FROM STUDENT
  WHERE ID<500
```

Cost =

STUDENT (ID, Name, LastName, Email, City, ...)



2.5K blocks, 150K tuples

```
SELECT *
  FROM STUDENT
  WHERE ID='54'
```

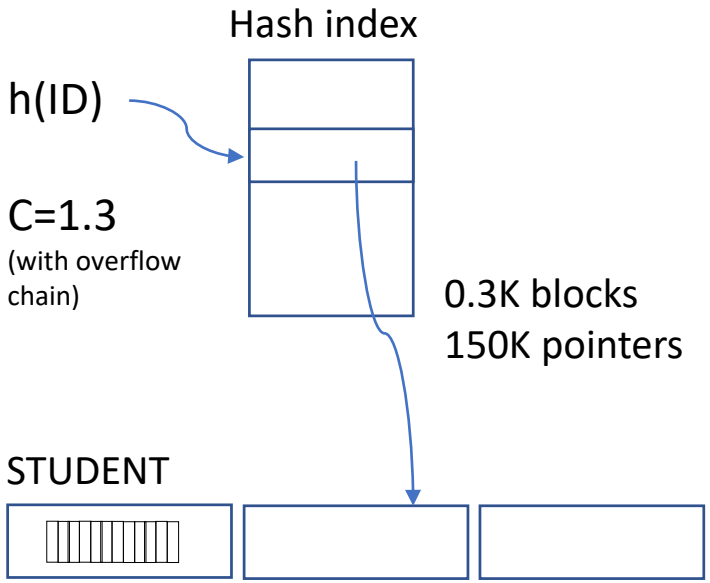
Cost =

```
SELECT *
  FROM STUDENT
  WHERE City="Milan
```

Cost =

```
SELECT *
  FROM STUDENT
  WHERE ID<500
```

Cost =




```
SELECT *
  FROM STUDENT
  WHERE City="Milan
```

Cost =

```
SELECT *
  FROM STUDENT
  WHERE ID<500
```

Cost =

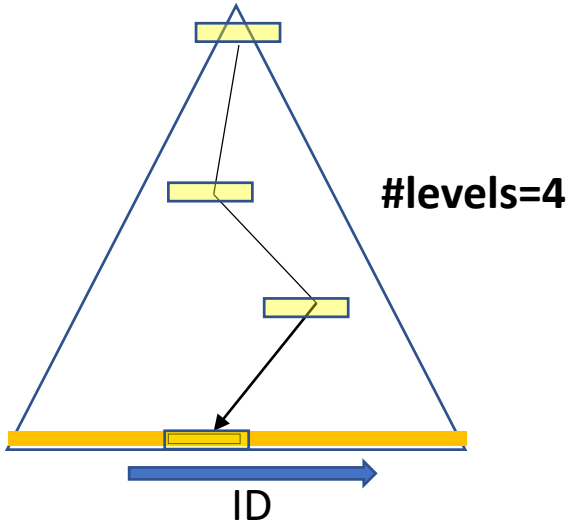
```
SELECT *
   FROM STUDENT
   WHERE LastName="Rossi"
```

Cost =

val(LastName)=75K

STUDENT (ID, Name, LastName, Email, City, ...)

B+ tree on ID



#leaf nodes (blocks) =2K

150K tuples

val(City) = 150

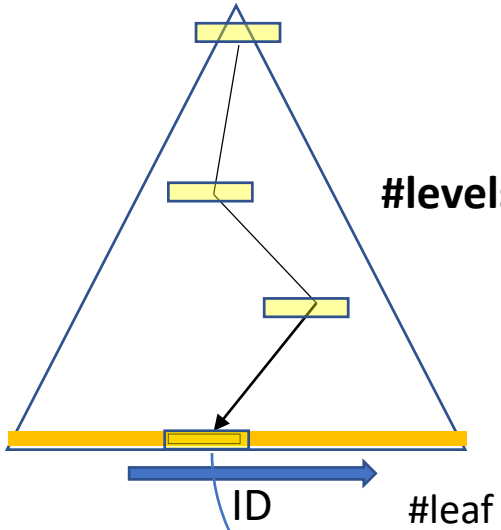
```
SELECT *
  FROM STUDENT
  WHERE ID='54'
```

Cost =

```
SELECT *
  FROM STUDENT
  WHERE City="Milan
```

Cost =

B+ index on ID



#levels=4

#leaf nodes (blocks) =0.5K
150K pointers

STUDENT (ID, Name, LastName, Email, City, ...)



1.7K blocks, 150K tuples

```
SELECT *
  FROM STUDENT
  WHERE ID='54'
```

Cost =

```
SELECT *
  FROM STUDENT
  WHERE City="Milan
```

Cost =

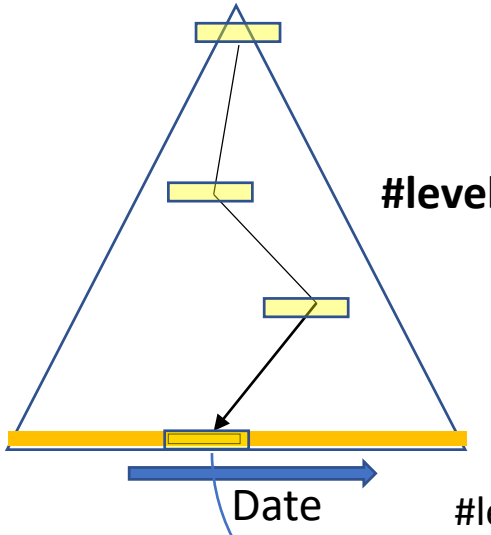
```
SELECT *
   FROM STUDENT
   WHERE LastName="Rossi"
```

Cost =

val(City) = 150

val(LastName)=75K

B+ index on Date



#levels=4

#leaf nodes (blocks) =2K
1.8M pointers

EXAM (SID, CourseID, Date, Grade)



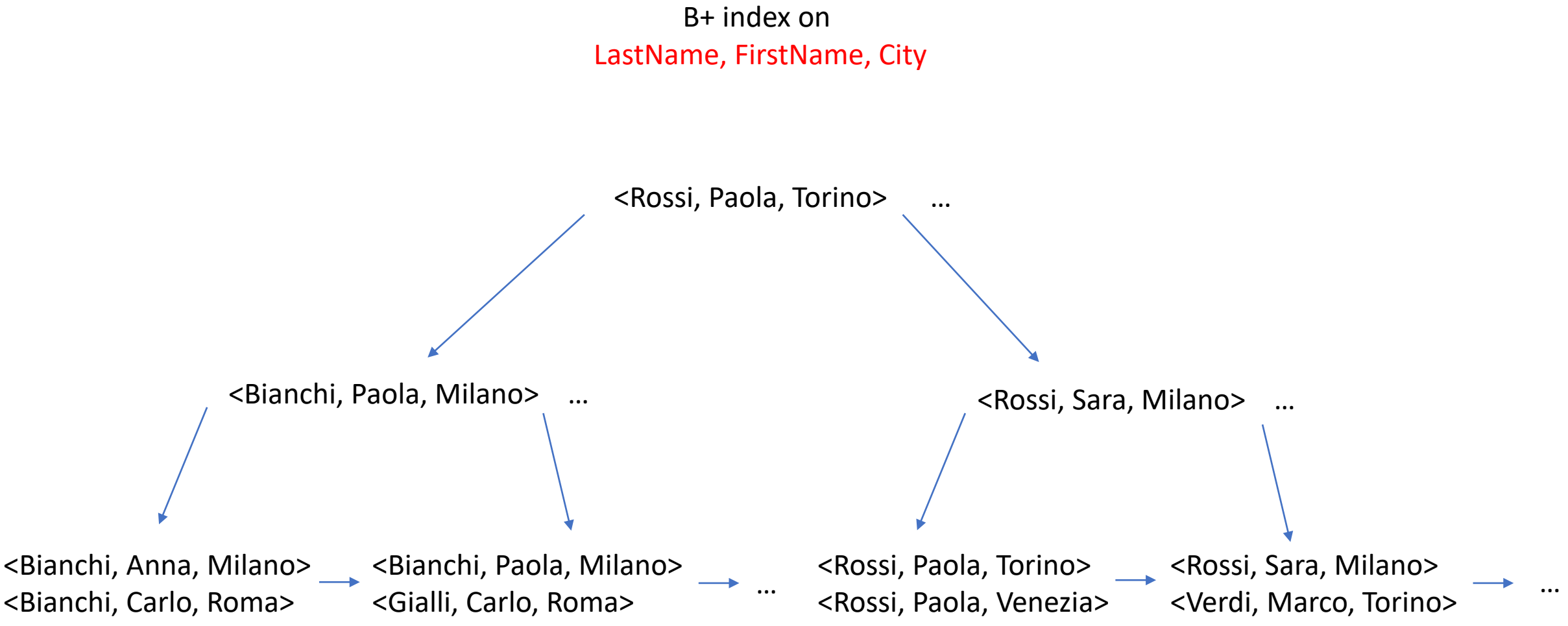
5.3K blocks, 1.8M tuples

val(Date) = 500

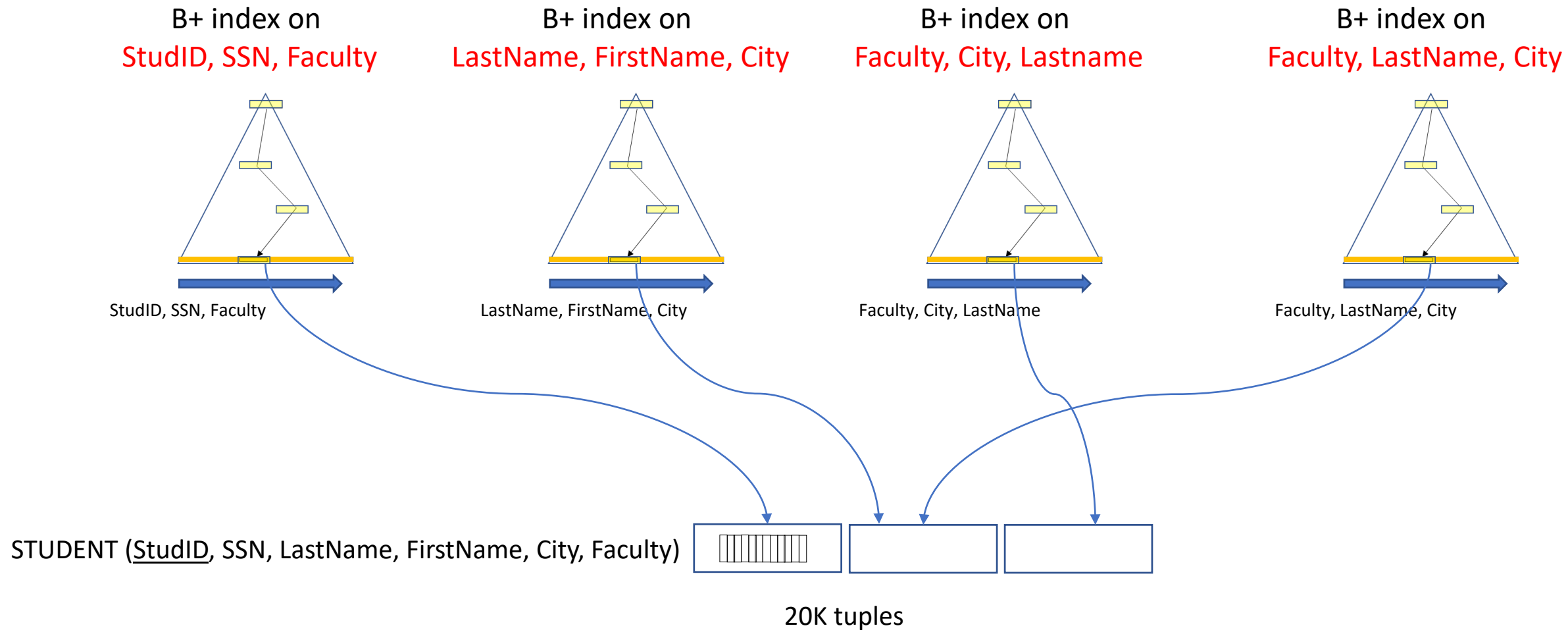
```
SELECT *
  FROM EXAM
  WHERE Date="10/6/2019"
```

Cost =

STUDENT (<u>StudID</u>, SSN, LastName, FirstName, City, Faculty)



[+ pointers to blocks with full tuples]



val(LastName) = 5000
val(FirstName) = 1000
val(City) = 1000
val(Faculty) = 10

```
select * from Student
where City = 'Milan' and
      Faculty = 'Computer Science'
```

STUDENT (ID, Name, LastName, Email, City, ...)



1.7K blocks, 150K tuples

EXAM (SID, CourseID, Date, Grade)



5.3K blocks, 1.8M tuples

```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
```

Nested loop join:

STUDENT (ID, Name, LastName, Email, City, ...)



1.7K blocks, 150K tuples

CITIES (City, Region, Description)



10 blocks, 1000 tuples

```
SELECT STUDENT.*
FROM STUDENT NATURAL JOIN CITIES
```

Nested loop join:

val(City) = 150

val(Grade) = 17

STUDENT (**ID**, Name, LastName, Email, **City**, ...)



1.7K blocks, 150K tuples

EXAM (**SID**, CourseID, Date, **Grade**)



5.3K blocks, 1.8M tuples

```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
WHERE City='Milan' AND Grade='30'
```

Scan STUDENT, lookup EXAM:

Scan EXAM, lookup STUDENT:

val(City) = 150

val(Grade) = 17

STUDENT (ID, Name, LastName, Email, City, ...)

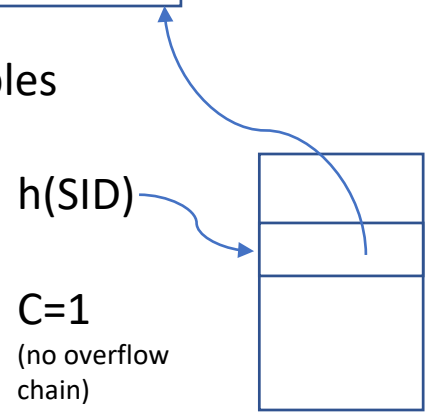


1.7K blocks, 150K tuples

EXAM (SID, CourseID, Date, Grade)



5.3K blocks, 1.8M tuples



3K blocks
1.8M pointers

```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
WHERE City='Milan' AND Grade='30'
```

Scan STUDENT, lookup EXAM:

```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
WHERE City='Milan' AND Grade='30'
```

Scan STUDENT, lookup EXAM:

STUDENT (ID, Name, LastName, Email, City, ...)



1.7K blocks, 150K tuples

EXAM (SID, CourseID, Date, Grade)



5.3K blocks, 1.8M tuples

```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
```

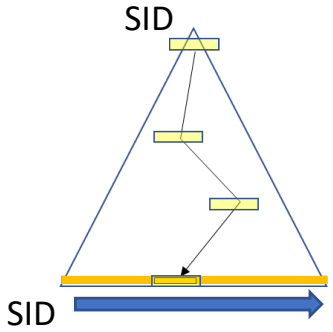
Merge scan join:

STUDENT (ID, Name, LastName, Email, City, ...)



1.7K blocks, 150K tuples

EXAM (SID, CourseID, Date, Grade)



8K blocks, 1.8M tuples

```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
```

Indexed merge scan join:

STUDENT (ID, Name, LastName, Email, City, ...)

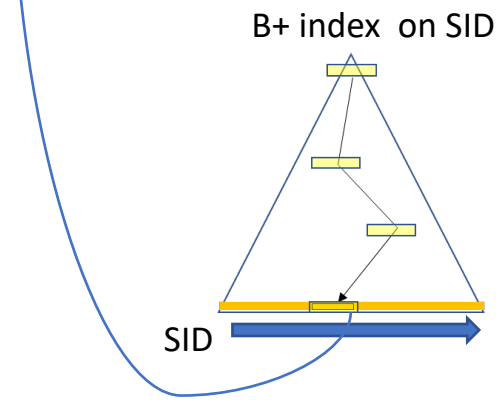


1.7K blocks, 150K tuples

EXAM (SID, CourseID, Date, Grade)



5.3K blocks, 1.8M tuples



4K blocks, 1.8M pointers

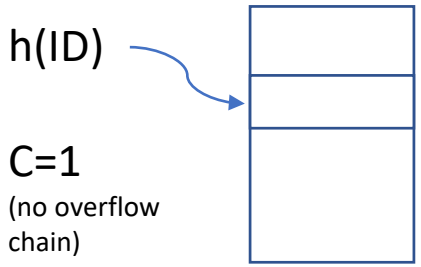
```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
```

Indexed merge scan join:

```
SELECT S.SID
FROM STUDENT S JOIN EXAM E ON ID=SID
WHERE GRADE>17
```

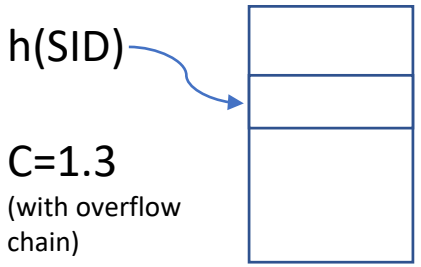
Indexed merge scan join:

STUDENT (ID, Name, LastName, Email, City, ...)



6K blocks, 150K tuples

EXAM (SID, CourseID, Date, Grade)

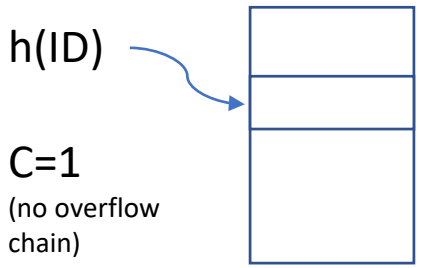


8K blocks, 1.8M tuples

```
SELECT STUDENT.*
FROM STUDENT JOIN EXAM ON ID=SID
```

Hash join:

STUDENT (ID, Name, LastName, Email, City, ...)

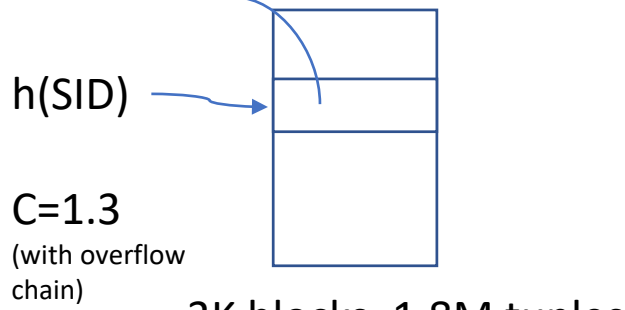


6K blocks, 150K tuples

EXAM (SID, CourseID, Date, Grade)



5.3K blocks, 1.8M tuples



3K blocks, 1.8M tuples